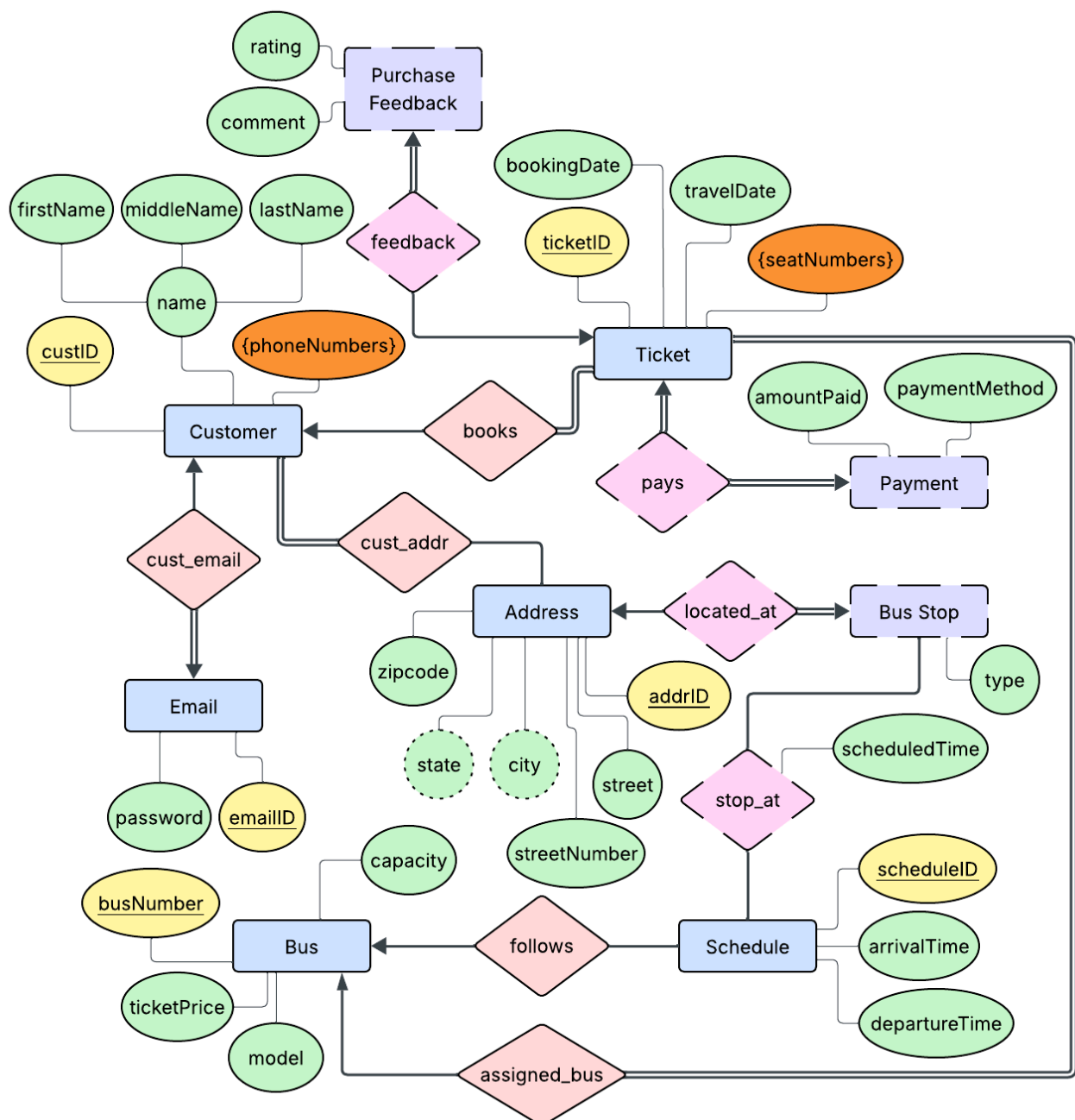


CSE 4701 - Homework 3

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Question 1



Due to the limitations of the program that I was using to make this, I could not use double outlined shapes. Weak entity sets are the purple boxes with dashed outline and the pink diamonds are the identifying relationships for weak entity sets.

Question 2

Assume that all attribute types are `int`.

1. Translate strong entities and un-nest composite attributes

```
create table A (  
    a1,  
    a11,  
    a12,  
    a2,  
    a3,  
    primary key (a1)  
);  
  
create table B(  
    b1,  
    b2,  
    primary key (b1, b2)  
);  
  
create table C(  
    c1,  
    c11,  
    c12,  
    c2,  
    primary key (c2)  
);  
  
create table E (  
    e1,  
    e2,  
    primary key (e1)  
);
```

2. Translate weak entities

```
-- other tables remain the same

create table D(
    d1,
    a1,
    c2,
    foreign key (c2) references C(c2),
    foreign key (a1) references A(a1)
    primary key (d1, a1, c2)
);
```

3. Translate multi-valued attributes

```
-- other tables remain the same

-- Multivalued attribute b3 for B
create table B_b3 (
    b1,
    b2,
    b3,
    primary key (b1, b2, b3),
    foreign key (b1, b2) references B(b1, b2)
);

-- Multivalued attribute d2 for D
create table D_d2 (
    d1,
    a1,
    c2,
    d2,
    primary key (d1, a1, c2, d2),
    foreign key (d1, a1, c2) references D(d1, a1, c2)
);
```

4. Translate relationships

```
create table A (  
    a1,  
    a11,  
    a12,  
    a2,  
    a3,  
    -- Relationship X  
    b1,  
    b2,  
    primary key (a1),  
    foreign key (b1, b2) references B(b1, b2)  
);  
  
create table B (  
    b1,  
    b2,  
    primary key (b1, b2)  
);  
  
create table C (  
    c1,  
    c11,  
    c12,  
    c2,  
    primary key (c2)  
);  
  
create table E (  
    e1,  
    e2,  
    primary key (e1)  
);  
  
-- Weak entity D (dependent on A and C)  
create table D (  
    d1,  
    a1,  
    c2,  
    primary key (d1, a1, c2),  
    foreign key (a1) references A(a1),  
    foreign key (c2) references C(c2)  
);
```

```
-- Multivalued attribute b3 for entity B
create table B_b3 (
    b1,
    b3,
    primary key (b1, b3),
    foreign key (b1) references B(b1)
);

-- Multivalued attribute d2 for weak entity D
create table D_d2 (
    d1,
    a1,
    c2,
    d2,
    primary key (d1, a1, c2, d2),
    foreign key (d1, a1, c2) references D(d1, a1, c2)
);

-- Relationship Z between B and E
create table Z (
    b1,
    b2,
    e1,
    c2,
    z1,
    primary key (b1, b2, c2, e1),
    foreign key (b1, b2) references B(b1, b2),
    foreign key (e1) references E(e1),
    foreign key (c2) references C(c2)
);
```

Question 3

1.

$C \rightarrow B$ and $B \rightarrow D$ gives $C \rightarrow BD$. When combined with $A \rightarrow BD$, we get $AC^+ = \{A, B, C, D\}$ which is a minimal candidate key.

R is in 1NF (R violates 2NF because of the partial dependency $C \rightarrow B$)

2.

Using each of the FD's, we can get $A \rightarrow BCD$. We can then add E to get the candidate key $AE^+ = \{A, B, C, D, E\}$.

R is in 1NF (R violates 2NF because B, C, and D all depend on A alone and A is a subset of the key).

3.

$A \rightarrow BC, A \rightarrow BCD, A \rightarrow BCDE$. So $A^+ = \{A, B, C, D, E\}$ is a candidate key.

R is in 2NF (R violates 3NF because D is not a key and E is not prime, but $D \rightarrow E$)

4.

$B \rightarrow C, B \rightarrow ACD, B \rightarrow ACDE$. So $B^+ = \{A, B, C, D, E\}$ is a candidate key.

R is in 2NF (R violates 3NF because A is not a key and C is not prime, but $A \rightarrow C$)

5.

$AC \rightarrow BE, AC \rightarrow BDE$. So $AC^+ = \{A, B, C, D, E\}$ is a candidate key.

$BC \rightarrow AD, BC \rightarrow ADE$. So $BC^+ = \{A, B, C, D, E\}$ is a candidate key.

R is in 1NF (R violates 2NF because of the partial dependency $A \rightarrow D$)

6.

$A \rightarrow BD, A \rightarrow BCD$. So $A^+ = \{A, B, C, D\}$ is a candidate key.

$B \rightarrow AC, B \rightarrow ACD$. So $B^+ = \{A, B, C, D\}$ is a candidate key.

$C \rightarrow B, C \rightarrow AB, C \rightarrow ABD$. So $C^+ = \{A, B, C, D\}$ is a candidate key.

R is in BCNF (all keys are single-attribute and all FD's have superkeys on the left side.)